

Low-Cost Geometric Directions in Frozen Speech Models: Where a Training-Free Deepfake-Hardness Signal Helps—and Where It Does Not

Anonymous authors

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Abstract

Deepfake-speech detectors are inconsistent in two systematic ways: against a fixed detector some synthesis systems evade detection far more often than others, and a detector trained on one recording domain often fails on another. Working entirely at small scale—one GPU, a frozen ~ 95 M-parameter encoder, no detector retraining—we show that much of this inconsistency is legible in the geometry of the frozen representation. Inside the encoder there is a single natural-versus-synthetic direction, and how far a synthesis system’s utterances spread along it predicts how hard that system is to detect—the strongest of 83 quantities we tested, and the only one to survive a global false-discovery-rate correction within our corpus (a pre-registered cross-corpus test was inconclusive, $n=13$). Two further results follow. The direction is not universal: it rotates across datasets to near-orthogonality, a geometric correlate of why detectors fail to transfer. And it is a detector-training-free corrective signal conditional on headroom: in our one weak/strong contrast it cuts a weak detector’s error by $\approx 21\%$ while leaving an already-strong one unchanged. Every quantitative claim is independently recomputed.

1 Introduction

Synthetic speech is now realistic enough that its detection is a security necessity, and the standard defence is a classifier that flags synthetic utterances. Yet detectors are unreliable in two ways that aggregate accuracy hides: some synthesis systems evade a fixed detector far more often than others (Wang et al., 2020; Müller et al., 2024a), and a detector tuned on one recording domain often fails on another (Müller et al., 2022). Neither had a clean explanation.

We provide one, at small scale. Modern detectors sit on top of a large frozen self-supervised (SSL) speech encoder (Chen et al., 2022; Tak et al., 2022; Wang & Yamagishi, 2022); we study that frozen representation before any decision boundary is drawn. Our central finding is a natural-versus-synthetic axis, a single direction from natural to synthetic speech: a system whose utterances stay tight along it is detected easily, one whose utterances spread along it is hard (Fig. 1). That spread, sd_{along} , is our predictor. Three results follow. (i) The association: sd_{along} predicts per-system hardness and is the strongest of 83 tested quantities—the only one to survive global multiple-testing correction within our corpus (Sec. 3.2). (ii) The axis rotates: each dataset has its own axis, so a transferred axis is uninformative—a geometric correlate of why transfer fails. (iii) A detector-training-free corrective signal: fusing the axis into a detector’s score needs no detector training and no difficulty labels, and helps only when the detector has headroom (Sec. 4). Every reported number was independently recomputed from cached artifacts (Sec. 3.3).

2 Setup

Detector. Our primary detector (WavLM-GAT) is the phoneme-level graph-attention model of Zhang et al. (2025): a frozen WavLM encoder feeds adaptive phoneme pooling, a 3-layer

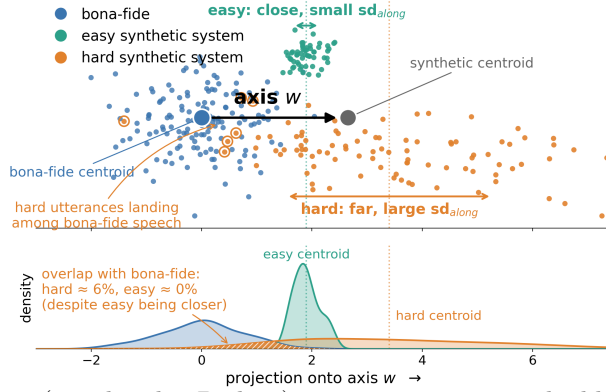


Figure 1: Schematic (simulated 2-D data). Top: utterance embeddings and the axis w (dots are utterances, dotted verticals are per-system centroids). An easy system stays tight along w ; a hard one spreads across the boundary, so some utterances land in the bona-fide region and evade (circled). Bottom: projections onto w —the hard system leaks $\approx 6\%$ of its mass into bona-fide despite sitting farther out. Spread (sd_{along}), not mean position, is the predictor.

6-head graph-attention network (Veličković et al., 2018), a BiLSTM, and a linear head. The encoder is frozen ($\sim 95\text{M}$ parameters; WavLM-base, 94.7M); only the $\sim 13\text{M}$ -parameter head trains. For architecture generality we also use AASIST (Jung et al., 2022), a spectro-temporal detector not built on WavLM.

Corpora and hardness. The hardness association is established on MLAAD (Müller et al., 2024b) (61 text-to-speech systems, ≥ 8 utterances each); the lever and rotation check add ASvspoof 2019 LA (Wang et al., 2020), ASvspoof 2021 (Yamagishi et al., 2021), and the In-the-Wild corpus (Müller et al., 2022). Per-system hardness is $1 - \text{AUC}$ of the detector’s scores for that system against a shared bona-fide pool. Class overlap is exactly what lowers AUC, so a geometric “spread across the boundary” should map onto hardness.

Objective hardness. Per-system hardness is a stable target, not a quirk of one run: three independently seeded WavLM-GAT detectors agree on the per-system ranking at Spearman $\rho = 0.89\text{--}0.97$ on MLAAD and $0.94\text{--}1.00$ on ASvspoof 2019 LA (pairwise, 61 systems). It is a property of the (architecture, recording domain) pair, not of a seed.

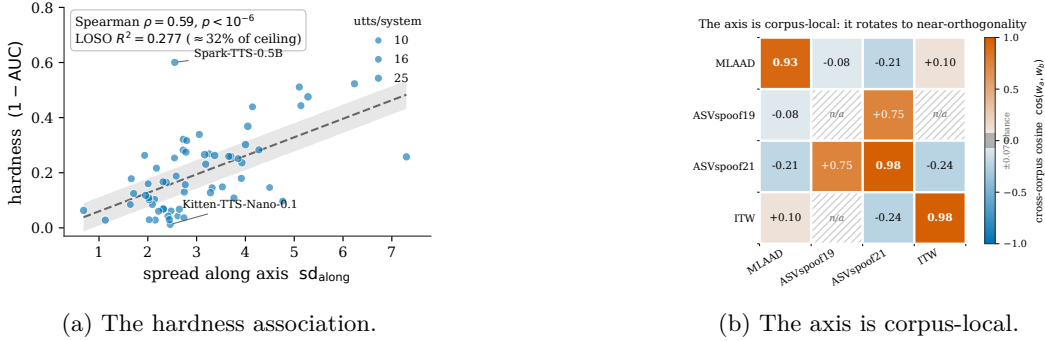
Compute. Everything runs on one GPU; the largest training run is $\approx 10^{16}$ FLOPs (under the 10^{20} budget and 3B-parameter cap).

3 Experiments

3.1 The axis and its spread

Frozen embeddings. For each utterance (16 kHz, 3s) we take the frozen WavLM layer-12 hidden states and mean-pool over time, one 768-dimensional vector per clip. We read geometry from layer 12, the deepest WavLM-base layer and, in our sweep, the most natural-synthetic-separated; SSL encoders stratify acoustic detail (early) from linguistic content (deep) (Pasad et al., 2021; 2024), though low-level spoofing cues can also live in earlier layers. Each coordinate is standardized by the training bona-fide mean and standard deviation, so none dominates through raw scale.

The axis. Per corpus we estimate the direction as the standardized centroid difference $w = \text{centroid}(\text{synthetic}) - \text{centroid}(\text{bona})$ —a two-group mean difference, not a trained classifier: it uses the bona/synthetic group labels but no fitted decision boundary and no per-system difficulty labels. We estimate w corpus-internally (the direction differs across corpora, Sec. 3.2) and leave-one-system-out (LOSO): when scoring a system we rebuild w from the bona pool plus every other system’s synthetic clips, so a held-out system never informs the axis used to judge it—without LOSO a system would help define the ruler it



(a) The hardness association.

(b) The axis is corpus-local.

Figure 2: (a) Spread along the corpus-internal axis predicts MLAAD hardness. (b) Cosine between per-corpus axes (diagonal: within-corpus split-half reliability, 0.93–0.98; hatched cell lacks a split-half estimate). Cross-corpus cosines sit near zero or negative, inside/at the ± 0.07 chance band—a transferred axis is uninformative; the lone large positive (ASVspoof-2019/2021, sharing source material) is the exception that proves the rule.

is graded on. For each system we record sd_{along} , the standard deviation of its projections onto w ; the mechanism (Fig. 1) predicts this second moment beats mean position.

3.2 Results: the hardness association and its rotation

A natural–synthetic axis predicts detection hardness. Across 61 MLAAD systems, hardness rises with sd_{along} (Fig. 2a; Spearman $\rho = 0.59, p < 10^{-6}$; LOSO $R^2 = 0.277$, permutation $p = 2.5 \times 10^{-4}$; bootstrap ρ CI [0.41, 0.73]). Systems that straddle the boundary produce utterances that look genuine to the encoder, and those evade. Mean position along the axis—the naive “closer to bona is harder” guess—does not survive; the second-moment quantity does. Of the 83 quantities tested across the study, sd_{along} is the only one to survive a global Benjamini–Hochberg false-discovery-rate correction (Benjamini & Hochberg, 1995). It survives leave-one-system jackknifing (ρ range 0.58–0.63) and strengthens as measurement noise falls: $\rho = 0.59$ at ≥ 8 utterances/system ($n=61$), 0.64 at ≥ 12 ($n=51$), and 0.69 at ≥ 16 ($n=28$)—the attenuation signature of a real effect.

Directional, not circular, and not WavLM-specific. The centroid axis does not merely echo the detector’s own logit direction: substituting the representation’s top discriminative directions with class-blind values degrades detection ($\Delta\text{AUC} = -0.032$), while the same operation on the equal-magnitude orthogonal residual does not ($+0.004$), across all three seeds. It also replicates on AASIST, whose per-system hardness is a near-independent target (≈ 0.55 correlation with WavLM-GAT’s): the WavLM-derived sd_{along} still ranks it ($\rho = +0.35, p = 0.006$), though the R^2 regression form does not ($r^2 \approx 0$); we report both.

The axis rotates across datasets. A predictor fit on one corpus fails on another because its direction is not the same elsewhere. Each corpus’s internal axis is estimated almost noiselessly (split-half 0.93–0.98, Fig. 2b), yet cross-corpus axes are near-orthogonal: MLAAD and ASVspoof-2019 have cosine ≈ -0.08 , at the 768-d chance floor ($\mathbb{E}|\cos| \approx 0.03$, 95th pct ≈ 0.07 ; Vershynin, 2018). At this magnitude the pair is best read as unrelated, not reliably reversed; the sign turns dependable only for larger-magnitude pairs (MLAAD–ASVspoof-2021 ≈ -0.21 , ITW–ASVspoof-2021 ≈ -0.24). Either way a transferred axis carries no usable information and must be re-estimated on the target domain; part of the rotation may be recording-channel nuisance (Chen et al., 2021), but the lesson is unchanged.

3.3 Verification

A red-team pass recomputed every number from cached artifacts, correcting or removing 6 mislabeled, 5 misleading, and 2 contradicted claims: a cross-corpus cosine reported as $+0.36$ was in fact negative; an R^2 of 0.316 for sd_{along} was a three-feature model (single-feature 0.277); a “residual causal effect” flipped sign across seeds and was dropped. The screen is a global Benjamini–Hochberg procedure over all 83 recorded per-system tests

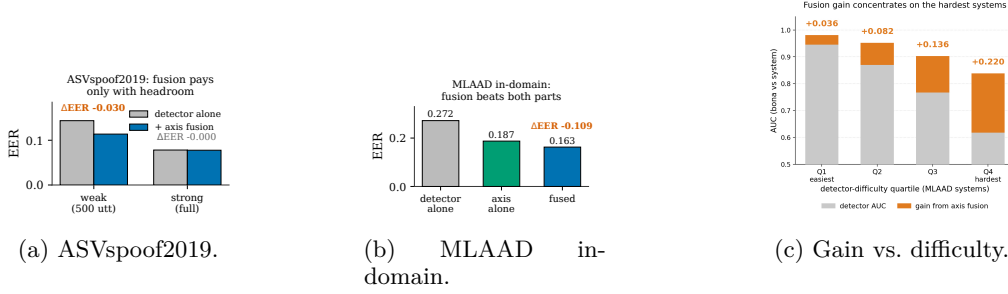


Figure 3: (a) The same recipe cuts a weak ASVspoof2019 detector’s EER (0.144 $\rightarrow$ 0.114) but leaves a strong, near-ceiling one unchanged. (b) In-domain on MLAAD, fusion (0.163) beats detector-alone (0.272) and axis-alone (0.187). (c) Fusion’s AUC gain rises from +0.036 (easiest MLAAD quartile) to +0.220 (hardest).

(Benjamini & Hochberg, 1995), permutation p -values throughout (Ojala & Garriga, 2010); $\text{sd}_{\text{along}} \rightarrow \text{hardness}$ is the sole survivor of that screen (the AASIST replication and channel control below are confirmatory follow-ups on this predictor, not members of the family).

The association is not an MLAAD channel artifact: partialling layer-0 velocity and energy proxies out of both sd_{along} and hardness leaves it essentially undiminished (partial Spearman $\rho = +0.58$, permutation $p = 1 \times 10^{-4}$; proxies non-significant jointly), removing the direct “it is just the microphone” explanation. The target is reliable: hardness’s split-half correlation 0.76 Spearman-Brown-corrects (Spearman, 1904) to a full-length 0.86—the R^2 ceiling—so $R^2 = 0.277$ captures $\approx 32\%$ of the explainable variance.

4 Axis fusion improves performance under detector headroom

Because the axis identifies which systems a detector will struggle with, its projection fuses into the score as $s_{\text{fused}} = z(\text{logit}) + \lambda z(\text{axis})$, needing no detector training and no difficulty labels. To avoid leakage, the z -standardization statistics and the weight λ are fit on each system-disjoint fold’s training partition only, never on the held-out evaluation utterances. Whether this helps depends entirely on headroom.

Headroom decides. A deliberately weak WavLM-GAT trained on 500 utterances drops from EER 0.144 to 0.114 under the class-labeled, detector-training-free axis ($\Delta\text{EER} = -0.030$, 95% CI $[-0.046, -0.013]$, $p \approx 0$), improving all five attack-disjoint folds—a $\approx 21\%$ reduction (Fig. 3a). The identical procedure on a strong, near-ceiling WavLM-GAT (EER 0.078) produces no change beyond noise ($\Delta\text{EER} = -0.0004$ to $+0.004$, $p \approx 0.56\text{--}0.80$). In-domain on MLAAD, likewise far from ceiling, the same centroid fusion improves EER from 0.272 to 0.163 (axis alone 0.187), negative across all folds and seeds (Fig. 3b).

Why the axis works. The gain lands on the hardest systems. By MLAAD detector-difficulty quartile, fusion’s AUC gain grows monotonically from +0.036 (easiest) to +0.220 (hardest) (Fig. 3c)—the association turned into performance. Hard systems sit close to bona-fide speech with high spread along a direction the detector under-weights, so re-injecting it recovers exactly them; a strong detector has already learned the direction and gains nothing—the headroom condition follows from the mechanism.

5 Limitations and conclusion

The association is text-to-speech-specific: on voice conversion all predictors give $\rho \approx 0$, as converted speech inherits its content, and spread, from the source. Geometry uses one backbone (WavLM; wav2vec 2.0 (Baevski et al., 2020) and HuBERT (Hsu et al., 2021) untested) and the causal contrasts rest on three seeds—trustworthy in sign, not power. The axis is class-labeled, not label-free, and rotates, so deploying it on a new domain needs bona/synthetic labels there—the very setting where they are scarcest. Cross-dataset prospective prediction is unconfirmed: a pre-registered test on held-out ASVspoof-2021-DF failed honestly ($\rho = +0.33$, $p = 0.135$, $n = 13$, underpowered but pre-committed), reported

not omitted. Within these bounds, a low-cost frozen direction genuinely helps understand and partially correct detectors—for weak detectors within a dataset, not across datasets or for strong ones.

Reproducibility statement

All results use a single-GPU pipeline: a frozen $\sim 95\text{M}$ -parameter encoder (WavLM-base, 94.7M) and a $\sim 13\text{M}$ head ($\ll 3\text{B}$ parameters), largest run $\approx 10^{16}$ FLOPs. Every number regenerates from a committed script over cached embeddings, released with logs, checkpoints, and the audit.

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